

Modified Schwarzschild Spacetime with Logarithmic Horizon Corrections

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Abstract. We solve the vacuum field equations for $f(R) = R^{(67/48)}$ gravity under static, spherical symmetry.

1 Introduction

Black holes are the primary laboratory for quantum gravity. We derive the modified Schwarzschild metric in SDGFT.

2 Theoretical Formulation

Here we formulate the core mathematical relations governing the system:

$$ds^2 = -A(r) dt^2 + \frac{dr^2}{B(r)} + r^2 d\Omega^2 \quad (1)$$

$$A(r) = 1 - \frac{2GM}{r} + \xi_{\text{LIV}} \ln\left(\frac{r}{2GM}\right) \quad (2)$$

3 Horizon Modifications

The logarithmic correction modifies the horizon structure and the Hawking temperature profile, preventing the black hole from evaporating completely.

4 Conclusion

The horizon modification resolves the information paradox by leaving a stable, planckian remnant that stores the information.

References

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